

General Industries Hub

Enterprise Inventory Management

Auditor Manual

Read-only review across every site: reports, records, KPIs.

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Confidential - for authorized personnel only

GI Hub - Role booklet - Auditor Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
1	Quality Control (QC)	One site or one warehouse - never both
2	Head of Department	One site
2	Head of Qualities (QC-HOD)	All sites, Surface Shield only, read + escalate
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

The Head of Qualities is level 2 but reads across every site, and that is not a contradiction. Cross-site reach normally comes from being level 3 - and level 3 would also have handed this role every endpoint the rank admits, which is most of the system. Instead it is level 2 with a named cross-site exemption, and the rank grants it nothing at all: a level check refuses it outright, so it reaches only what names the role explicitly. Its whole surface is the Quality Oversight page. See §23.

The QC role is scoped on one of two axes, and exactly one. Material is inspected either where it arrives (a warehouse) or where it lands (a site), and which one depends on how your operation is organised. A QC account is therefore created against a site or against a warehouse, and an account carrying neither - or both - sees nothing at all rather than everything. That is deliberate: an inspector with an unclear remit is a bigger problem than an inspector with none.

2.2 Page access matrix - which role can open which page

[!] A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Read that caveat together with the table below it, not instead of it: the caveat says a page is locked to "its own role", and the table is what names that role. A Head of Department can open Man-Hours and the Material Estimator; those two are locked to the HOD, not away from them. Admin is the single exception and reaches every workspace deliberately, for support.

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Live Dashboard	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Stock	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[X]	[OK]	[OK]
Lots & Expiry (\$3.10.3)	[OK]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[X]	[OK]
Rack Locator	[OK]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Assets	[OK]	[OK]	[OK]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Entry Log (all tabs)	[OK]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[X]	[X]	[OK] (shadow)
HOD Portal -> Approvals	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
HOD Portal -> Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Lining Coverage	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK]	[OK]
Bulk	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Excel Import									
WBS & Work Types	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Material Estimator (SME)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (read)	[OK]
Man-Hours / Manpower Tracking	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Execution Entries	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[X]	[OK]
Reports	[X]	[X]	[X]	[X]	[OK]	[OK]	[X]	[OK] (read + download)	[OK]
Logistics Portal (Procurement)	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Warehouse Portal (Receiving & DN)	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK] (shadow)
Quality -> Inspections	[OK] (read)	[OK] (read)	[X]	[OK] (decides)	[OK] (read)	[OK] (read)	[X]	[OK] (read)	[OK]
Quality -> QC Accounts	[X]	[OK]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Quality Oversight (\$23)	[X]	[X]	[X]	[X]	[X]	[X]	[OK]	[X]	[OK]
Safety & People -> PPE Forecast	[OK]	[X]	[X]	[X]	[OK]	[OK]	[X]	[X]	[OK]
Safety & People -> PPE Usable Time	[OK]	[X]	[X]	[X]	[OK]	[X]	[X]	[X]	[OK]
Safety & People -> Employees	[OK]	[X]	[OK]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Records -> Inventory	[OK]	[OK]	[OK]	[OK]	[OK] (+ New item / Edit, own site)	[OK]	[OK]	[OK] (read)	[OK]
Records -> Receipts, Consumption, Returns, Lots, Purchase Requests	[X]	[X]	[X]	[X]	[OK]	[OK]	[OK]	[OK] (read)	[OK]
Records -> Purchases	[X]	[OK]	[X]	[X]	[X]	[OK]	[X]	[OK] (read)	[OK]

Page	Store Keeper	Warehouse User	Supervisor	QC	HOD	Logistics	QC-HOD	Auditor	Admin
Issue Orders									
Valuation & 30-Day Burn (\$24.3)	[X]	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]	[OK]
Admin Portal	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Documents - Security - Training - Feedback	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]

Where each role lands after signing in: Store Keeper -> Entry Log -> Issue Stock - Warehouse User -> Warehouse Portal - Supervisor -> Supervisor Portal - QC -> Quality -> Inspections - HOD -> HOD Portal -> Approvals - Logistics -> Logistics Portal - Head of Qualities -> Quality Oversight - Auditor -> Dashboard - Admin -> Admin Portal -> Console.

2.3 What each role can do

One list per role, in plain words. If you only want to know whether your account can do a thing, read your own entry - each is complete on its own.

2.3.1 Store Keeper - what a Store Keeper can and cannot do

Can: open the Entry Log and post every stock movement - receive, issue, return, adjust, count, returnable items, OCR import; see Incoming Deliveries and Supervisor Requests; open the Dashboard, Stock, Rack Locator, Assets, PPE Forecast, PPE Usable Time, Employees and Execution Entries; open Lots & Expiry and Records -> Inventory; read quality inspections; download documents.

Cannot: approve anything (an HOD approves what a Store Keeper stages); open the HOD, Supervisor, Logistics or Warehouse portals; open Reports, the ledger Records (Receipts, Consumption, Returns, Lots), the Material Estimator or Man-Hours; see another site's stock.

Scope: one site.

2.3.2 Supervisor - what a Supervisor can and cannot do

Can: raise material requests from the Supervisor Portal; post Execution Entries; open the Dashboard, Stock, Assets and Employees.

Cannot: post stock movements (the Entry Log is the Store Keeper's); approve requests; open Reports, the ledger Records, Lots & Expiry, the Material Estimator, Man-Hours or any other role's portal.

Scope: one site.

2.3.3 Quality Control (QC) - what an inspector can and cannot do

Can: decide inspections - pass, fail or hold - from Quality -> Inspections; open the Stock page to see what is waiting and what their decision released; open Lots & Expiry to see each lot's expiry and what is left of it (§3.10.3).

Cannot: open the Dashboard, post any stock movement, approve a purchase, create QC accounts, or move themselves between sites (that is an Admin decision). See §22.

Scope: exactly one site or exactly one warehouse - never both, never neither.

2.3.4 Head of Department (HOD) - what an HOD can and cannot do

Can - and this is the question people ask most: open the Man-Hours / Manpower Tracking page and the Material Estimator. Both are exact-locked to the HOD (and Admin for support); the lock is what admits an HOD, not what excludes them. That includes all thirteen Man-Hours tabs - roster, timesheets, estimator, variance, scorecard, employee-wise, the four execution views, the Manpower Planner and the SME Session Plan (§19).

Also: approve or reject everything the site stages (receipts, issues, returns, adjustments, counts); create and submit Purchase Requisitions to Logistics; raise Cross-Site Requests; run Reports; open the Executive Summary, Burn Rate, Low Stock, Lining Coverage, Document Library and Bulk Excel Import; create QC accounts for their site; post Execution Entries; open PPE Forecast, PPE Usable Time and Employees; open Lots & Expiry (§3.10.3); add and edit inventory items for their own site from Records -> Inventory (deleting stays Admin-only).

Cannot: open the Logistics Portal, the Warehouse Portal, the Supervisor Portal, the Entry Log, Quality Oversight or the Admin Portal; raise a Purchase Order (Logistics does that); see another site's stock - though they may request material from one.

Scope: one site.

2.3.5 Head of Qualities (QC-HOD) - what this role can and cannot do

Can: open the Quality Oversight page and, since Phase 16, Lots & Expiry (§3.10.3) - every lot's expiry and what is left. Across every site and warehouse, but only for Surface Shield material: the controlled category is filtered in the database on every query, so it is the boundary of the role rather than a filter on a page. Seven views - overview, Surface Shield purchase orders, the MTC register, where it is used, stagnation and expiry, escalations, and the 90/60-day thresholds. They may raise and close escalations and retune those two thresholds.

Cannot: approve or reject a quality inspection; move, issue or receive stock; raise a PR or PO; open any other portal or page; see PPE, tools, consumables or any other category. See §23.

Scope: all sites and all warehouses; Surface Shield only.

2.3.6 Logistics Coordinator - what Logistics can and cannot do

Can: run the whole procurement chain from the Logistics Portal - receive submitted PRs, raise Purchase Orders, assign POs to warehouses, manage vendors, track DN status; open Records (inventory, ledgers, POs, PRs), Reports, the Warehouse Portal, Lining Coverage, the Rack Locator, PPE Forecast, and quality inspections; create QC accounts.

Cannot: open the HOD Portal, the Entry Log, the Supervisor Portal, the Material Estimator, Man-Hours or the Admin Portal - outranking a role does not open its workspace.

Scope: all sites, no site lock.

2.3.7 Warehouse Operator - what a Warehouse User can and cannot do

Can: receive against POs, prepare and dispatch Delivery Notes from the Warehouse Portal; open the Dashboard, Stock, Rack Locator and Assets; read quality inspections; create QC accounts for their warehouse.

Cannot: open the Logistics Portal, the HOD Portal, the Entry Log, Reports, Records, the Material Estimator or Man-Hours.

Scope: one warehouse.

2.3.8 Auditor - what a view-only Auditor can and cannot do

Can: read across every site - Dashboard, Stock, Assets, Records, Reports (including downloads), Lining Coverage, the Material Estimator, the HOD Portal's read-only views (Executive Summary, Burn Rate, Documents, Low Stock, PRs, Cross-Site Requests), Employees and quality inspections.

Cannot: change anything at all, anywhere. Every write route is refused by the server, not merely hidden. Not Man-Hours, not the Entry Log, not any approval. See §20.

Scope: all sites, read-only.

2.3.9 Administrator - what an Admin can do

Everything, including every other role's workspace as a support shadow, plus the Admin Portal: users, access requests, overdue actions, inventory master, the audit log and the console.

Adding an inventory item (Admin -> Inventory Master -> New item). Pick the Site and the Category from the suggestions. The server matches them to the spelling already in use, ignoring capitals and a plural s, so cncec is saved as CNCEC and consumable as Consumables. This matters because a store keeper only sees items whose site matches theirs exactly. If you type a site or category that no item uses yet, GI Hub asks "Create a new site...?" first. Press Cancel if you meant an existing one. A new item must have a site.

The HOD adds and edits items too (30 September 2026). On the Inventory page (Records -> Inventory), the Admin and the HOD have a New item button and an Edit button on each row. The same form, the same Site / Category matching and the same audit log apply. The HOD's items are always saved to their

own site, which the form shows instead of asking. A HOD cannot add or edit another site's items. Deleting an item stays Admin-only (Admin -> Inventory), because it cannot be undone.

2.4 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Dashboard, Stock, Material Requests and Execution Entries are all site-locked.
Quality Control (QC)	Whichever single place they were created against: one site, or one warehouse. A site QC sees inspections raised at that site and has no warehouse business at all; a warehouse QC is pinned to its warehouse exactly as a Warehouse User is. An account with no binding, or with both, sees an empty list - it never falls back to seeing everything. Moving a QC to another site is a request, decided by an Admin, not something the QC or their HOD can do alone.
HOD	Their own site only - but they can REQUEST material from other sites (Cross-Site tab) and submit PRs to Logistics.
Logistics	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Head of Qualities	All sites globally, but ONLY Surface Shield material - every query on their page is filtered to the controlled category in the database, so the category is the boundary of the role rather than a filter on a page. They read, and they send escalations; they cannot approve an inspection, move stock, raise a PR or open any other portal. See §23.
Auditor	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It can open the Dashboard, Stock, Records, Reports and Lining Coverage and change nothing anywhere. See §20.
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar when shadowing the Warehouse Portal.

A site-locked user is never asked for their site (30 September 2026). On Issue, Receive, Return, Adjust, the supervisor's material request and the HOD's purchase request, a Store Keeper, Supervisor or HOD sees their site as a blue label (CNCEC your site), and the entry is saved there. The same goes for the

"All sites" filters on Approvals, Burn Rate, Reports, SME and the Executive Summary. Picking another site used to be possible and only produced an error, because the server allows your own site only. An account with no site assigned still sees the dropdown; ask an Admin to set the site.

2.5 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics, warehouse_user, qc, qc_hod or auditor accounts are seeded. All of them are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Live | Practice switch - above the username. Live is the real system and is selected unless you choose otherwise. Practice is a separate training copy with made-up data, where nothing you do affects Live. Switching signs you out. See §26 Practice Mode for everything about it.
- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open. This works from any address, including a bookmarked /login; an address that is not a page takes you to your home page instead of a blank screen.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.1.1 The look of the sign-in page

Phase 14e, redesigned in Phase 15c.

The large gold GI mark sits in its own space at the top of the page, and the glass sign-in card sits below it, so the two never overlap. The mark rises into place once when the page opens. After that it stands still, and a slow band of light sweeps across the gold every few seconds. Nothing on the page follows the mouse. What you see depends on your device, and the form works the same on all of them, from the first moment:

- Everyone, including the Android, iOS and desktop apps: the mark drawn in the page's style.
- A desktop computer with a graphics card: the same mark in 3D gold, in the same place. It loads separately after the page is ready, so it never delays signing in. It draws nothing between sweeps, stops when you switch tabs, and is removed as soon as you sign in.
- A very short window (under about 560 pixels tall): the large mark gives way to a small, still mark inside the card.
- Anyone with Reduce motion switched on in their system settings: no rise and no sweep. The mark simply stands there.

The Executive Summary and the Training page use the same glass style. Their cards rise gently into place as you scroll, except with Reduce motion on and when printing.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

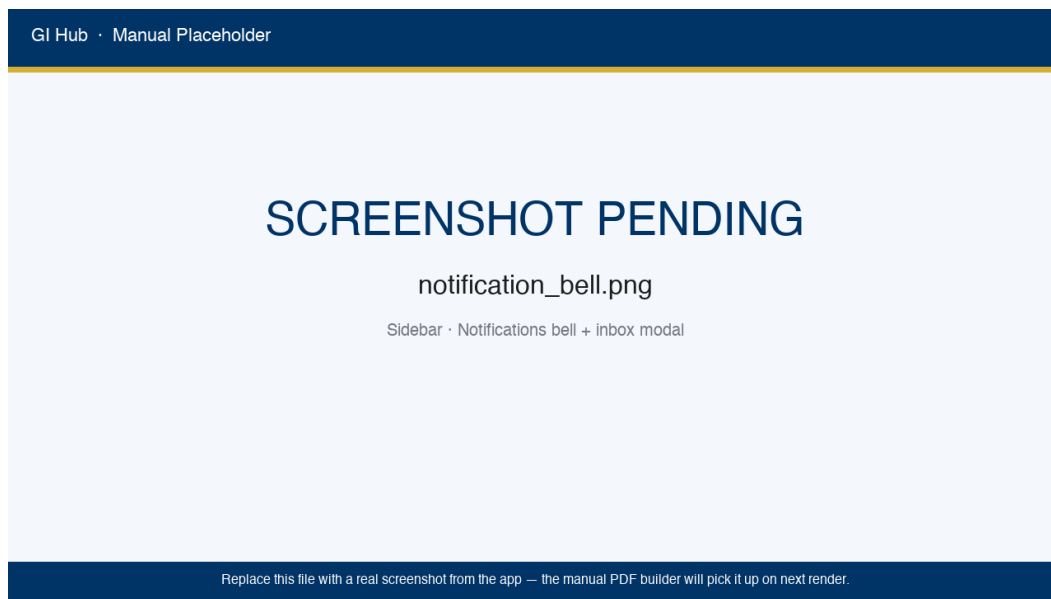
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

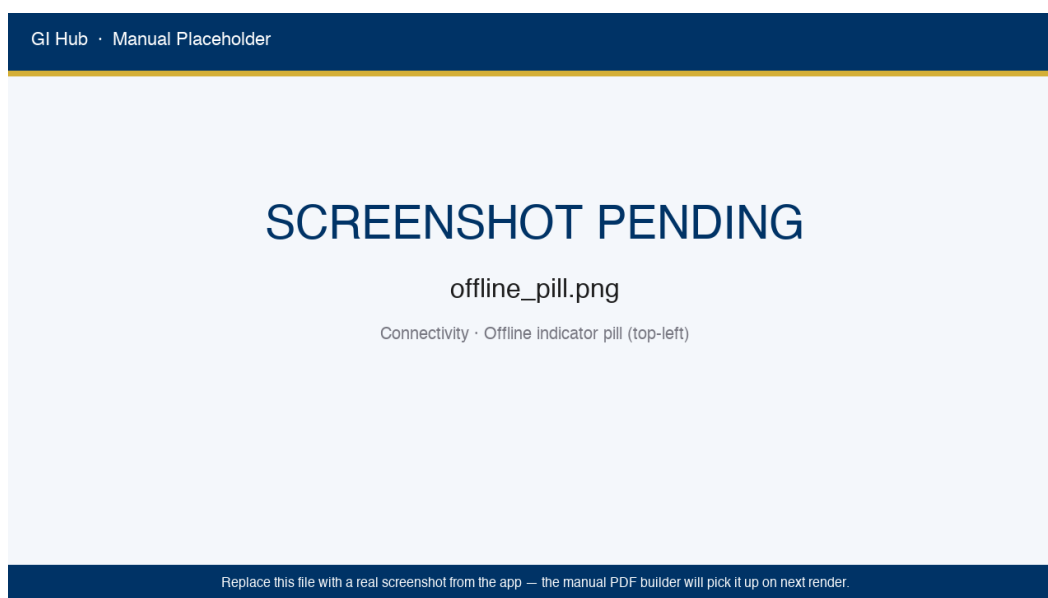
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line

Event you'll see	Triggered by
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

3.7 Saving an entry when the signal drops

If the network drops while you save a receipt, issue, return, stock adjustment or bulk batch, GI Hub keeps the entry on your device and shows a saved offline message instead of an error. A badge in the header shows how many entries are waiting. They are sent by themselves when the connection returns, and you can press the badge to send them straight away.

An entry is never saved twice. Each save carries its own reference, and the server recognises it if it arrives again. That can happen when the page reloads while the entry is being sent, or when the connection drops just as the server confirms. The second arrival is matched to the first, and only one entry reaches your HOD. You never need to check for, or delete, a duplicate caused by a bad connection.

3.8 What's new

Phase 14d.

When a feature you use changes, a What's new panel opens by itself the next time you sign in. It shows only what concerns you: an announcement is sent to the roles that can open the page it is about, so nobody is told about a screen they cannot reach.

Each card says what changed. Open it takes you to the page, Watch the tutorial plays the short video when there is one, and a User Manual § tag names the section to read.

Press Got it and the panel closes. It does not open by itself again for the same announcements. To see recent ones, press the gift icon in the top bar, next to the bell; a number on it means something new is waiting.

A matching notification also appears in the bell (§3.6). Announcements are in-app only - never WhatsApp or email.

3.9 Stock vs the Excel workbook - which materials differ, and why

Added 26 September 2026.

GI Hub keeps its own stock ledger, and the site also keeps CNCEC_Inventory.xlsx. After every Excel sync, GI Hub compares its stock for every material with the workbook's Current Stock column. It then works out why any material differs.

Where: the top of the Stock page.

- A yellow banner says how many materials differ, for example 14 of 505 materials don't match the Excel workbook, and when that was checked.
- Each of those materials has a red row and a red != Excel tag. Point at the tag to see the reason.
- Review the differences lists every material. Each one shows its workbook and GI Hub figures, what caused the difference, where to look (a GI Hub record number or an Excel sheet and row) and how to fix it.
- A material's own page (scan its QR or open it from Stock) shows the same warning.

What the causes mean:

Tag	What happened	How to fix it
Only in GI Hub	A receipt, issue or return was entered in GI Hub, and the workbook's log never got it. The entry names who made it and its remarks.	If it really happened, add it to the log. If it was a test or a mistake, reverse it in GI Hub with a Stock Adjustment, or have an admin remove it.
No longer in Excel	GI Hub took a line from the workbook, and the workbook has since lost it (deleted or re-dated).	Put the line back if that was a mistake. Otherwise an admin removes it with the sync's --prune-vanished option.
Not synced yet	The workbook has a line GI Hub doesn't have yet.	Run the Excel sync.
Edited in Excel	A quantity was changed in the workbook after the last sync.	Run the Excel sync.
Sheet totals stale	The Inventory sheet's own Current Stock doesn't match its own logs. Current Stock is a typed value in this workbook, so it can fall behind.	Correct the figure on the Inventory sheet.
Opening stock	The Opening Stock differs.	Run the sync, or correct the workbook.
Unexplained	Part of the difference that none of the above explains. It is always shown, never hidden.	Compare the material's records by hand.

The marked workbook. Press Get the marked workbook and choose your CNCEC_Inventory.xlsx. You get back a copy containing:

- a GI Hub check sheet first, listing every cause with how to fix it;
- each differing material's row on the Inventory sheet filled red, with a note on its Current Stock cell;
- any log line the check names filled amber.

Make the fixes in your own workbook, then sync again. Your file itself is never changed. Check again uploads the workbook and refreshes the list without downloading anything.

Store Keepers, Warehouse, HODs, Logistics and Admins can upload. Everyone who can see Stock can read the result. Nothing on this screen changes stock.

[!] The check only explains. It never corrects stock by itself, because deciding whether an entry was real is a person's call.

3.10 Lots, FEFO and the Lot Register workbook

Phase 16 (October 2026).

A lot is one batch of a material: the cans of PU COMP A from batch 3504, or the CHEMOLINE rolls of batch 1O25003382. GI Hub keeps the stock of each lot, so it can tell the store keeper which lot to use first (FEFO, first expire, first out).

3.10.1 What the `Serial No.` column means in the workbooks

The Receipt, Consumption and Return Logs have one Serial No. column. GI Hub reads it according to the item on that row:

The item is...	Serial No. is...	GI Hub keeps it as...
a Surface Shield (PU, Furan, Phenacin, Coroflake, BC 3004...)	the batch	the Lot No. (and the cell as typed)
a roll (CHEMOLINE, UOM ROL)	the roll number	the roll number, and its batch as the Lot No.
a brick (AR 30/40 MM, carbon bricks)	not a lot - bricks have no batch and never expire	the serial, with no lot
anything else (pumps, tools, equipment)	the equipment / asset no.	the serial, as before

Rules that keep this safe:

- A lot is spelled one way. 3504.0 is 3504, and A4525 and A 4525 are the same lot (A 4525). N/A or a blank cell means no lot.
- Roll numbers begin with 1O (the letter O). A roll typed 10... (with a zero) is corrected to 1O..., and the sync says so.
- One lot per cell. A cell such as 4525 = 55 Cans; 1823 = 2 Cans gets no lot. Split it into one row per lot in the workbook; the sync names the rows that need it.
- A lot that was never received is reported, not invented. If the Consumption Log names a lot no receipt brought in for that material, the sync lists it with the material it was received under (often a typing slip), and the consumption still counts toward stock.

- History keeps the lot it was given. GI Hub never picks a lot for an old workbook row that has none. Automatic FEFO applies only to issues entered in the app.
- A return gives back to its lot. The Return Log's Serial No. is now read too: a Surface Shield return carries its lot, and an equipment return keeps its asset tag.

Every receipt that names a lot creates that lot in GI Hub. A lot's stock is always received - consumed - returned, worked out from the ledger.

3.10.2 The Lot Register workbook ('Rubber & Brick Materials - CNCEC.xlsx')

This workbook keeps, per Surface Shield material, one sheet of what arrived: batch, manufacture date (MFD), expiry, DN No. and quantity. Put it in the same folder as the other workbooks. The normal sync reads it automatically; any file named like *Rubber*Brick*CNCEC*.xlsx is found, and the run prints a > lots section.

[!] It never changes stock. Stock comes only from the Receipt, Consumption and Return Logs. This workbook only describes the lots GI Hub already has: their MFD, expiry and batch. Its quantities are compared with the Receipt Log, and every difference is reported. A mistake in it can make an expiry wrong, but it can never double a quantity.

How each row is matched:

1. To its material - by the SAP column. If a sheet has no SAP column, GI Hub matches the material name, the component letter (COMP C) and the pack size (10 Kg vs 2.5 Kg), and refuses anything it cannot pin to exactly one item. The material code is only a cross-check, because COMP A-D share one code.
2. To its receipt - the same SAP, DN No. and received date. CHEMOLINE rolls have no DN, so they match on SAP and date, roll by roll.
3. To its lot - the Batch No. (normalised as in §3.10.1). For CHEMOLINE the Order No. is the batch and each Roll No. is a roll inside it.

What it fills in:

On...	It writes...
the lot	MFD, expiry, batch reference, DN No.
the roll register	each CHEMOLINE roll and the batch it belongs to
a receipt with no lot	the lot, when the workbook names exactly one batch for it
an item with no shelf life	the shelf life (months), learned from its own MFD -> expiry pairs

Expiry:

- An expiry in the workbook is used as it is (from the file).
- A missing expiry is MFD + the item's shelf life (derived).
- If two rows of one lot disagree, the earliest expiry is used.
- An expiry before its MFD is refused and reported.
- An expiry a person typed on the Receive form is never overwritten.

Bricks are compared with the receipts but get no lot (they have no batch and never expire).

What the sync tells you (each is listed by sheet and row):

- quantities that differ from the Receipt Log;
- rows with no matching receipt;
- expiries before the MFD;
- lots whose rows disagree on dates;
- material codes that disagree with the inventory;
- a batch the Receipt Log spells differently (e.g. 0926 vs 0.926);
- rows with no batch.

3.10.3 Lots & Expiry - which lot to use, and what is about to expire

The Lots & Expiry page (sidebar, under Stock) is for the Store Keeper, QC, the HOD and the Head of Qualities. It lists every lot of a lot-tracked material, the oldest expiry first, with:

- MFD and expiry - an expiry marked derived is MFD + the item's shelf life;
- status - Expired, <= 30 days, <= 60 days, <= 90 days, OK, No expiry, or Quarantined / Disposed when the Admin has taken the lot out of use (Admin Console -> Lots). A quarantined or disposed lot is never offered on the Issue form;
- received, consumed, returned and remaining - worked out from the ledger.

Click a status at the top to show only those lots. Show used-up lots adds the lots with nothing left. A CHEMOLINE batch opens to show its rolls: when each was received, where it is, and for a used roll the day and tank it was issued to. A red banner warns when an expired lot still has stock. At the bottom, Lots used but never received lists lots the Consumption or Return Log names that no receipt brought in.

Issuing (the Issue form's Lot field). For a lot-tracked material, the Lot field is a list of that material's open lots in FEFO order. Each shows its expiry, the days left and the quantity left.

- The first one is marked FEFO. Leaving the field blank issues from it, because GI Hub's own automatic pick uses the same order.
- [!] An expired lot is listed last, marked red, and is never the suggestion, even though its date is the earliest. You can still choose it; FEFO warns, it never blocks.
- Choosing any lot other than the FEFO lot asks for a reason, which goes to the HOD with the issue. Choosing the FEFO lot asks nothing.
- CHEMOLINE: pick the roll; its batch fills in by itself. A roll already issued is no longer offered.

Receiving (the Receive form). For a lot-tracked material the form asks for the batch (as printed on the label), the manufacture date (MFD) and the expiry:

- If you leave the expiry blank, it becomes MFD + the item's shelf life, and the form shows the date it will use.
- An expiry before the MFD, or an MFD in the future, is refused.
- An expiry you type is kept, and the Lot Register workbook never overwrites it.

The daily expiry notice. Each evening the Store Keeper and the HOD of a site get one notice listing that site's lots that expire within 30 days or have already expired, each with how much is left. It is one notice per site, never one per lot. On WhatsApp it arrives inside the evening summary.

Shelf life and lot tracking (item editor). Admin -> Inventory or Records -> Inventory -> Edit has two

settings:

- Lot tracking: Automatic - every Surface Shield except bricks - or Tracked by lot, or Not tracked.
- Shelf life (months): the number a missing expiry is derived from. The Lot Register workbook fills it in where it is empty, from the item's own MFD -> expiry pairs (PU 9, Phenacin 6, BC 3004 24...).

Reading the two warnings - they are different things.

You see...	It means...	What to do
a red Expired lot with stock left	The lot was received, its expiry has passed, and the ledger says some of it is still on the shelf.	Check the shelf. If it is there, ask the Admin to dispose of it or quarantine it. If it is not, an issue or return was never entered - enter it, or correct the workbook.
a lot under Lots used but never received	The Consumption or Return Log names a lot that no receipt of that material brought in. The lot itself may be fine; usually the lot number or the SAP code on that line was typed wrong. The column Received under SAP shows the material that did receive a lot with that number.	Correct the line in the workbook (the lot, or the SAP code) and sync again. The stock already counts the line either way.

An expired lot whose stock is all used shows Used up, not Expired, and appears only with Show used-up lots switched on. So when a lot shows received 1, consumed 1, remaining 0, there is nothing to dispose of.

3.10.4 Practising with lots

Practice (\$26) has lots to try all of this on:

- PRACTICE PU PRIMER (899971) has three lots: PR-OLD (expired, still has stock), PR-SOON (expires within 30 days, so it is the FEFO suggestion) and PR-LATE.
- PRACTICE CHEMOLINE (899973) has batch 1O26009999 with three rolls to pick.
- One issue names lot PR-TYPO, which no receipt brought in, so the Lots & Expiry page has a used but never received example.

3.11 What changed in September-October 2026 (Phases 14, 15 and 16)

A one-page list for everyone who used GI Hub before September 2026. Each line names who it affects and where it is explained in full.

3.11.1 Phase 14 - quantities, reconciliation and announcements (26 September)

Change	Who	Read
Surface Shield quantities are shown both ways, kilograms first (189 KG - 21 Can). You still enter packs; a consumption form may be written in KG and is converted.	Store Keeper, Supervisor, HOD	§4.11, §4.10.8
When a QR form and the Excel workbook both record the same drum, the ledger keeps one quantity (the larger), never both.	HOD	§6.17
The Surface Shield queue is grouped: one job = one tank on one day, area asked once and credited once.	Store Keeper, HOD	§4.9a.2
What's new notices after an update, and tutorial freshness warnings for admins.	Everyone; Admin	§3.8, §7.14
The new 3D login (a still crest on phones and older computers).	Everyone	§3.1
Stock vs the Excel workbook: the Stock page marks materials whose stock differs from the workbook's, and says why.	Store Keeper, HOD, Admin	§3.9
An entry saved while the signal dropped is sent once, however many times it is retried.	Store Keeper	§3.7

3.11.2 Phase 15 - Practice, Garnet and the store keeper's note (28-30 September)

Change	Who	Read
Practice is violet with a pulsing badge, and refuses to start on an out-of-date database (no more half-working pages).	Everyone	§26.2, §17.9
Garnet is surface preparation: it credits no area of its own; the HOD answers Old or New surface per job.	Store Keeper, HOD	§4.9a.7
The store keeper's note fills the job card: Floor - 13.37 SQM Done fills the area, the part and the remark. Two notes on one tank and day are two jobs. The remark goes to the	Store Keeper, HOD	§4.9a.2

Change	Who	Read
HOD, the job history and the exports.		
The HOD adds and edits inventory items for their own site (Records -> Inventory -> New item / Edit). Deleting stays Admin-only.	HOD	§2.3.4, §2.3.8-2.3.9 (Adding an inventory item)
A user who belongs to one site no longer sees a Site box on the forms: their site is filled in and shown, not asked. Admins and multi-site users still choose.	Store Keeper, HOD	this section
The WBS No. field appears on Receive and Issue. If the HOD has not set up any WBS numbers yet it shows "None set up" and the entry still saves.	Store Keeper, HOD	§13.14, §16.5
Site and Category on a new item are matched to the spelling already in use (cncec -> CNCEC).	Admin, HOD	§2.3.8-2.3.9 (Adding an inventory item)

3.11.3 Phase 16 - lots, FEFO and the Lot Register workbook (1 October)

Change	Who	Read
The workbooks' Serial No. is read by the item: the batch for a Surface Shield, the roll for CHEMOLINE, the asset tag for equipment. Live now has real lots.	Everyone	§3.10.1
The Lot Register workbook (Rubber & Brick Materials - CNCEC.xlsx) adds each lot's MFD, expiry and rolls. It never changes stock.	Admin	§3.10.2
Lots & Expiry page: every lot, its expiry, and what is left of it.	Store Keeper, QC, HOD, Head of Qualities	§3.10.3
The Issue form's Lot field lists lots in FEFO order; an expired lot is listed last and never suggested. Choosing another lot asks for a reason (it never blocks).	Store Keeper	§3.10.3
The Receive form asks for the MFD; a blank expiry becomes MFD + shelf life.	Store Keeper	§3.10.3

Change	Who	Read
One evening expiry notice per site lists lots expiring within 30 days.	Store Keeper, HOD	§3.10.3
A return gives back to its lot, so a lot's balance is received - consumed - returned.	Everyone	§3.10.1
Practice has example lots: an expired one, a FEFO one, CHEMOLINE rolls, and a lot used but never received.	Everyone (training)	§3.10.4

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

The header reads "Reports & Analytics". Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply remaining per item	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first,	Answering "what is on our shelves"

Report	What it shows	Typical use
	with the top-ten share	worth?"
Shelf-Life / Expiry	Lots grouped into expired, critical and warning	The weekly walk-round to pull short-dated stock
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first, and where they did not	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Builds the report for the type, date window and site you chose. The page shows a loading shimmer while it works.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	The report type and site, above the date range covered
v Download button	Saves the report in the format you picked - PDF, Excel or CSV
Summary cards	The headline figures for the report, filled in automatically
Bar chart	Shown for the Burn Rate and Daily Consumption reports
Preview table	The rows themselves, with status colours

8.2.5 A note about apostrophes in downloaded files

Sometimes a downloaded Excel or CSV file shows an apostrophe in front of a value you typed - for example '=A1 where you typed =A1. Nothing is wrong, and nothing has been lost.

Spreadsheet programs treat anything starting with an equals sign, a plus, a minus or an at-sign as a calculation to run, not as words to display. If that text came from something a person typed into the system, opening the file could make the spreadsheet run instructions nobody intended. The apostrophe tells the spreadsheet "this is text, show it exactly as it is".

- It only ever appears in front of text that begins with one of those symbols.
- Numbers are never affected. Quantities, prices and totals stay numbers, and every total in the file still adds up correctly.
- Ordinary words, notes and material names are untouched.
- If you need the plain text back, delete the leading apostrophe in the cell.

You will see this most often on notes and remarks fields, since those are the places people type freely.

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st 06:00
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label
- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

Available when AI is switched on and the local AI service is running:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 standard checks and rewrites the AI commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	How sure the AI is of this finding
Right-side metric callout	Headline number with sub-label
Body paragraph	The AI's explanation, written from the figures the check returned
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 standard checks:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If the local AI service is not running, the tab explains what an Admin needs to start.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
 - Re-email
 - Delete (audit-logged)
-

9. Automated Notifications - WhatsApp & Email & In-app Bell

The system automatically queues messages on key events. There are now THREE notification surfaces:

1. WhatsApp - messages queue and are sent by a background service, so a WhatsApp outage delays them but never loses them. There is a master on/off switch plus a per-event switch, so you can silence WhatsApp for one kind of event while leaving the in-app bell untouched for it.
2. In-app notifications bell (NEW in v3.0) - every event is also written to your personal inbox and surfaced by the sidebar bell described in §3.6. It ALWAYS fires, regardless of the WhatsApp toggle.
3. Email - the same Outlook / Mail app / mail-link and mail-server paths as before.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock count	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot + reason
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items result
HOD cross-site request created	All Admins	Submit cart	Includes target site + count
HOD cross-site request, > 5 items	Target Site HOD	Submit cart	Bulk escalation
Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack and ship"
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	Includes rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration	All Admins	Registration form	Includes username + role +

Event	Recipient	Trigger location	Notes
request			site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Recorded in the audit log as a manual WhatsApp
PR submitted to Logistics (v3.0)	Logistics role	HOD PR tab -> Submit PR(s) to Logistics	On by default
PO issued (v3.0)	Site HOD	Logistics -> Save PO	On by default
PO assigned to Warehouse (v3.0)	Warehouse users at that WH	Logistics -> Assign	On by default
Warehouse acknowledged (v3.0)	Logistics	Warehouse -> [OK] Acknowledge	Off by default - low value, kept quiet
Warehouse received goods (v3.0)	Logistics	Warehouse -> Record receipt	On by default
DN logistics approved (v3.0)	Site HOD	Logistics -> [OK] Approve DN	On by default
DN HOD approved -> SK (v3.0)	Site SK	HOD -> DN Approvals -> [OK] Approve	On by default; shares the DN-generated toggle
DN received at site (v3.0)	Logistics + Warehouse	SK -> [OK] Mark as Received	Off by default
Reschedule requested (v3.0)	Logistics	Warehouse / HOD -> Request reschedule	On by default
Reschedule decided (v3.0)	Requester	Logistics -> [OK] Approve / [X] Reject	On by default
Vendor return raised (v3.0)	Logistics	Any role -> <- Raise return	On by default
PR force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	On by default; critical severity
PO force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	On by default; critical severity
Delivery reminder T-2 / T-1 / T-0 (v3.0)	Logistics + HOD + Warehouse (per DN)	Daily sweep - see §9.3	Each of the three lead times has its own toggle

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> Draft Outlook Email	Mac: opens the Mail app with a formatted table. Windows: opens

Event	Trigger location	Mechanism
		Outlook with a formatted table. If neither is available, it falls back to a plain-text mail link
EOD report email	Sent by the mail server	The mail server address and sign-in details are set up once by an Admin
Report delivery (scheduled)	Reports -> Scheduled	Uses the same mail server

9.3 In-app notifications bell (NEW in v3.0)

See §3.6 for the UI. Every notification is stored against the recipient and read back scoped to their role, their site and their warehouse - you only ever see your own. Each event carries a severity - info, success, warning or critical - and that is what colours the left border of the card.

9.4 Delivery reminder daily sweep (NEW in v3.0)

Once a day the system looks ahead at upcoming deliveries and sends reminders two days before, one day before, and on the day itself:

Watched	When	Severity
A Purchase Order's expected delivery date	T-2 / T-1 / T-0	warning / warning / critical
A Delivery Note's delivery date	T-2 / T-1 / T-0	warning / warning / critical

PO reminders ping: Logistics + originating Site HOD. DN reminders ping: Logistics + Site HOD + Warehouse user(s) at the receiving warehouse.

No duplicates - the sweep cannot fire twice for the same delivery on the same day. Two guards:

1. Every reminder sent is recorded against the document, its delivery date and how many days out it was, and the same combination can only ever be recorded once.
2. The system also stamps the date it last ran the sweep, so once it has run for the day the check is skipped entirely rather than repeated every minute.

Restarting the reminder service mid-day is safe. Re-running the sweep manually on the same day sends zero new notifications.

Customising the cadence - the two-day, one-day and same-day pattern is currently fixed. Making it configurable per site is on the backlog; ask if you need a different rhythm.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (the unique identifier), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current Stock} = \text{Total Received} - \text{Total Consumed} - \text{Total Returned}$. It is never stored as a number anyone can type over; it is recalculated from the ledger every time it is shown.

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	Access tokens for the offline-capable mobile app
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered -> closed / force_closed / cancelled
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered / returned / closed / force_closed
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed / cancelled
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned / cancelled
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled

Table	What it stores	State machine
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	Unread -> read, stamped with the time it was read
delivery_reminders_sent	Record of which T-2 / T-1 / T-0 reminders have gone out	Write-once - the same reminder can never be recorded twice

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant GI-PS01, Delivery_Date Day 3.
Day 1 - 8:15 AM	Site HOD	Opens the Submit PR(s) to Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	A "PR submitted to Logistics" notification lands in the Logistics inbox with a red badge, plus a WhatsApp ping if that event is enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL. Clicks Use this PR to create a PO.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line, Total_Price computed. Save PO.
Day 1 - 9:30 AM	(auto)	The PR lines are marked as covered by a PO and disappear from the Logistics queue. The site HOD gets a "PO issued" notification with the

When	Who	What
		vendor and PO number.
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.
Day 1 - 10:00 AM	(auto)	A "PO assigned to warehouse" notification lands in the WH-A users' inbox, plus a WhatsApp ping if that event is enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt. Both lines are marked delivered.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for RL only.
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line -> DN-WH-A-20260617-002.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	Both DNs move to awaiting-HOD-approval. The site HOD gets the "DN approved by Logistics" notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via DN Approvals tab. Approves both.

When	Who	What
Day 2 - 6:00 PM	(auto)	Both DNs move to awaiting-Store-Keeper, and a matching draft receipt appears in the site's Receipt Staging for each. The site SK gets the incoming-DN notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.
Day 3 - 8:45 AM	(auto)	Two receipts are posted to the ledger, each carrying its DN number, source warehouse and originating PO. Both DNs are marked received, the staged drafts are cleared, and the Live Dashboard shows the new stock immediately.

16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 is marked rejected, and the reason is stored with it
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed

- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

16.4 What stops a duplicate PR or a double order

Added 2026-08 (Phase 8). Three rules run underneath the procurement chain. None of them changes how the screens are used; they change what happens when two people, or one impatient person, do the same thing twice.

16.4.1 A PR number is reserved, not guessed

Every PR number is claimed in a registry before the requisition's lines are written, so two HODs pressing Create PR in the same second get different numbers. If a number is somehow taken, the system takes the next one rather than failing. Renaming a PR moves its reservation with it and is refused if the new name is already in use - by a registered PR or by one that was imported without ever being registered.

16.4.2 Each step happens once, in order

A requisition moves site_draft -> submitted -> in_po, and each move is checked against the state the row is actually in - not the state the screen was showing when it loaded. Consequences you will see:

- Submitting twice does nothing the second time. It used to accept lines that were already submitted, rewrite their timestamp and fire a second notification at Logistics.
- A PO cannot be raised twice over the same lines. The second attempt is refused with a message naming the lines, rather than silently succeeding.
- A PR may still carry several POs. Partial fulfilment splits one requisition across vendors or deliveries, so the lock is per line, not per PR.

16.4.3 A retry is not a second order

Submit PR, Create PO, Assign PO and Rename PR each carry a key minted when the form opens. Press the button twice, lose the connection and retry, or double-click - the system recognises the repeat and replays the first answer instead of doing the work again. Sending the same key with different details is refused, because that is a client bug rather than a retry.

[!] The buttons disappear rather than grey out. Once a PR is submitted, the Submit button is gone - not disabled. A greyed-out button invites a second click and a support call about why it "isn't working"; an

absent one says the step is done.

20. Auditor (View-Only) Manual

Added 2026-08-03.

19.6 Efficiency by Day

Where: Man-Hours -> Efficiency by Day.

Answers the question the rest of the portal cannot: for the same lining system, how much manpower went into Equipment A against Equipment B?

19.6.1 Why man-hours per m², and not hours

A 400 m² tank will always show more hours than a 40 m² vessel. That says nothing about how well either was done. Man-hours per square metre is the only figure on this page that survives a change of size, which is what makes two jobs comparable at all.

The cards at the top give you that number per job, best first. The chart is there to show you how it got there.

19.6.2 [!] The line is the RUNNING figure, not the day's

The bars are the hours booked each day. The line is cumulative - all the hours so far divided by all the area so far.

That matters, and it is the easiest thing on the page to misread. A day-by-day figure swings far too hard to follow, and on a day with no area it does not exist at all. The running figure settles as a job goes on, and it is the number you would quote in a meeting.

It also means setup is included. A tank that reads 6.6 MH/m² after a fortnight of scaffolding is telling you what the job really cost. Its good days might read 2.2 on their own - that is the same tank, with the setup left out.

19.6.3 [!] Days with hours and no area

Mobilisation, scaffolding, curing and inspection all book hours against zero square metres. On those days:

- the line breaks. There is no area to divide by, so there is no figure. A zero would read as "this crew achieved nothing per metre", which is a claim about them rather than about the day.
- the hours still count towards the running figure. They are part of what the job cost.
- the day is listed underneath, with whatever the timekeeper wrote in the Remarks box on the timesheet.

[!] Where nothing was written, the table says "no reason recorded". The app does not guess at "scaffolding" - only the person who was there can say, and a guess in that column would become the record.

19.6.4 Comparing across lining systems

If your selection covers more than one lining system, the page warns you. A tile lining and a coat are different work with different benchmarks, so their man-hours per m² are not comparable - filter to one system to compare equipment against equipment.

It still draws the chart. You may have asked for exactly that view, and refusing to show it would teach nothing.

19.6.5 If the chart is empty

It fills in from two places: the Daily Timesheet (hours) and the team SQM recorded against a day's work. A site that has entered neither will see a sentence saying so, not a broken chart.

20.1 What the Auditor role is for

The Auditor is for anyone who must see the numbers without being able to move them - an internal auditor, a client-side QA reviewer, a finance reviewer, a new manager still learning the system, or a consultant given temporary access.

It is a real role like any other, with one difference that is absolute:

> An Auditor can read everything its level reaches and can change nothing, > anywhere, ever.

That is not a UI convention. It is enforced on the server for every request.

20.2 What an Auditor can see

Area	What it shows
Dashboard	Every KPI, chart and category breakdown, across all sites
Stock	Current stock by site, minimum levels, expiry
Records	Inventory, Receipts, Consumption, Returns, Lots, Purchase Orders, Purchase Requests
Reports	Every report, downloadable as Excel, PDF or CSV
Lining Coverage	The read-only coverage analysis
Documents	The SOP and User Manual PDFs, QR/label sheets
Account	Own security settings (password-manager 2FA, phone number)

An Auditor is not site-locked. Every other role below Logistics sees only its own site; an auditor sees all of them, because an auditor confined to one site cannot audit the business.

20.3 What an Auditor cannot do

Everything that changes data. The Data Entry forms, Master Data editors, Bulk Excel Import, HOD

Approvals, the Logistics and Warehouse portals, the Supervisor portal and the Admin console do not appear in the sidebar at all - they are not hidden buttons on a visible page, the pages are simply not reachable.

On pages an Auditor can open, actions that would change something are disabled rather than hidden, with a tooltip explaining why:

> Your account is view-only (Auditor) - this action changes data.

On the Reports page, for example, Excel / PDF / CSV downloads work normally (they only read), while Archive, WhatsApp, New schedule, Run now and Delete are greyed out.

A "View only" tag sits permanently in the header beside the user name, so there is never any doubt about which kind of account is signed in.

20.4 What an Auditor CAN still do

Because they are read-only, not crippled:

- Download every report in all three formats.
- Use the CmdK / Ctrl-K command palette to jump to any page or material.
- Ask the Hub Assistant questions (it answers from the Auditor's own manual chapters - see §20.6).
- Manage their own account: enable or disable two-factor authentication and change their own phone number. These touch only their own user record.

20.5 The security model (for Admins)

Worth understanding before you hand the role out.

The restriction is enforced once, centrally, rather than screen by screen. Any request that would change data from a view-only account is refused before it reaches the feature at all, and the user is told plainly that their account is view-only and the action is not permitted.

Why that design matters to you: a permission check added screen by screen is only as reliable as the person who remembers to add it, and the one that gets forgotten fails open - the change succeeds and nothing looks wrong. Checking centrally inverts that, so a feature added next year is closed to Auditors from the moment it is written. 126 of the system's 143 data-changing actions are blocked; the 17 that are not are sign-in, sign-out, staying signed in, the user's own two-factor and phone-change steps, and a handful of read-only calculations that are technically submissions only because the figures are too large to fit in a web address.

Two consequences to be aware of:

- An Auditor probing an address that does not exist is told "not permitted" rather than "not found". The check runs first, so it cannot be used to map the system.
- Blocking is by role, not by page. Even if an Auditor somehow reached a write page, every button on it would still fail server-side.

20.6 Creating an Auditor account

Admin Portal -> Users -> Add user, and pick Auditor (view-only) in the Role dropdown. Leave the Site blank - Auditors are global by design, and the form does not ask for one.

Users can also request the role themselves on the registration screen; it still needs Admin approval like any other request.

20.7 The Hub Assistant for Auditors

The assistant answers from chapters 1, 2, 3, 8, 9, 10, 11, 12, 16 and 20 - orientation, the reports module, notifications, the data model, the glossary, the FAQ and the cross-role procurement walk-through. That is deliberately the set matching the pages an Auditor can open.

Ask it about a report, a status code or what a field means and it will answer. Ask it how to post a consumption entry and it will tell you that is outside the read-only Auditor view - because the operational chapters for roles that can post entries are never loaded into an Auditor's context in the first place.

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: numbered comment lines inside the worked examples in the Operations chapter were being mistaken for chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore procedure. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose

What the label carries	What you get
Only the item code, and several units exist	A "which one is in your hand?" list to pick from
Only the item code, and no units are registered	The ordinary material view - the system does not invent an asset

Recording where something is. Open the item, tap Move, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

21.12 Phase 9 - what changed for you

Five things changed at once, and one of them changes how you work every day.

21.12.1 [!] Consumption is filed from paper now

This replaces the store-keeper-first flow entirely. Full detail in §4.10.

You fill a printed form -> Store Keeper verifies -> HOD approves

- Print a form per lining system (§4.9). It lists every material for that system, so nobody writes a material name by hand - that is the one thing the reader is genuinely bad at.
- Fill in the date, equipment and area at the top, and a quantity and a lot number on every row.
- Photograph the whole page including the QR code and upload it.
- Check every figure. Where the handwriting was not certain the box is left empty rather than guessed, and those rows are marked.
- The store keeper verifies your quantities next, not the HOD. Their corrections show to the HOD in red, and you are told before it is approved.

[!] Approval now deducts the material as well as posting the area. Store keepers: stop raising a separate material issue for lining work - the execution entry is the only writer now, and doing both deducts the same drum twice. (From Phase 14 a drum that is also logged in the Excel Consumption Log is reconciled rather than deducted twice - see §6.17 - but the rule stands: one record per drum.)

[!] Rejection is final. Raise a new entry from a fresh form.

[!] Each printed sheet is used once, and a sheet printed before its system's materials changed will be refused. Both rules exist because your handwriting is matched to materials by row position.

21.12.2 WBS numbers, at last

The WBS # column was blank on every entry in the system - not because the feature was missing, but because there was no screen to reach it. There is one now: HOD -> WBS & Work Types (§16.5).

- Add your WBS numbers, then map each work type to one.
- An issue that leaves the WBS box blank gets the number its work type maps to.
- Work types are now a fixed list you curate, which is what stops civil and Civil becoming two cost centres.
- Nothing is enforced until you add the first row. Turning it on is your act.
- WBS is applied forward only; historical entries are not restamped.

21.12.3 A new chart: Efficiency by Day

Man-Hours -> Efficiency by Day (§19.6). For one lining system, how much manpower went into Equipment A against Equipment B - normalised on man-hours per m², which is the only figure that survives a change of size.

[!] The line is the running figure, not the day's, and it breaks on a day with hours and no area. Both are deliberate; §19.6 explains why.

21.12.4 The Manpower Planner tells you what nights actually buy

Three changes to the arithmetic:

- Nights buy calendar time, not a smaller payroll. The total headcount is unchanged - nobody works both shifts - but the plan now shows how many days the night crew saves.
- The day/night split follows your roster, not a division by two. This site runs about 20 on days against 80 on nights; halving the requirement understated the night crew fourfold.
- Hiring idle trades no longer flatters the plan: capacity counts only the roles the job needs.

21.12.5 "Labor" is now "Manpower"

The page, the menu entry and every heading now say Manpower - the word everyone here actually uses.

> If you have automation reading the API: the field names did not > change. Done_SQM_Labor and Labor_Variance_Pct are still spelled exactly > as they were. Only what you see on screen moved, so nothing you have built > against those keys needs touching.

21.13 Photo reading - the two failures that are fixed

If you tried to read a photographed Daily Consumption log or the new printed consumption form and it did not work, this is what was happening and what has changed. Nothing about how you use it is different - you still photograph the page and wait.

21.13.1 "The log came back empty, or said it could not be read"

The reader was being cut off part-way through a long sheet. It read your page correctly, got as far as about row 13 or 14 of a 30-row log, and ran out of room - and because the answer was incomplete, the system threw away all of it, including the thirteen rows it had read perfectly.

This is also why Delivery Notes always worked: a note is four or five items and always finished comfortably. It was never that the reader could not handle your handwriting or the free-form table.

Now: it is given enough room for a full 30-row sheet, and if a page is ever still too long, the rows it did read are kept and only the unfinished one is dropped. On the operator's own test photo this went from nothing at all to all 30 rows.

21.13.2 "The new PDF just hangs, then fails"

A full page of handwriting genuinely takes the reader several minutes. The system was giving up after four - often within seconds of it finishing - and then telling you "the vision model is not reachable", which sent people off to check a service that was running perfectly.

Now: it waits long enough for a real page, and if something genuinely is wrong it says which of the two it is: still working, or actually unreachable.

21.13.3 Two smaller things you may notice

- A quantity the reader could not make out now arrives BLANK, not as 0. It used to show a confident zero, which reads like a number somebody wrote and nobody re-checks. A blank box is a question, and the row will not submit until you answer it.
- A photo that produced nothing now says so. Previously a failed read could finish looking merely "empty", which is indistinguishable from a photo of a blank form. If the page could not be read, you are told to retake it.

21.13.4 If a photo still will not read

Nothing has changed about the advice, but it is worth repeating: get the whole page in frame, keep the lighting even, and hold the camera square to the paper. The Paste tab works with the AI switched off

entirely and is always available. Typing the entry in by hand is never wrong.

21.13.5 [OK] Ditto marks now come through (2026-09-02)

Where: Entry -> OCR Import (/entry/ocr), the Consumption log lane.

If you have been retyping the Name, Tank No. and Product Name columns after every photo read, that was a real fault and it is fixed.

Most people write a ditto mark - " or (ditto) - instead of repeating themselves down a column, and on a typical sheet that is most of the page. The reader was seeing those marks perfectly well and quietly returning nothing for them, because to a reader "same as above" is not a word to transcribe. The step that was supposed to copy the value down was looking for the mark, found an empty cell instead, and left it empty. On the sheet this was diagnosed against, that was 19 of 30 tank numbers, 14 of 30 names and 8 of 30 product names.

Now: a dittoed cell is filled in from the row above, whether the reader returns the mark or returns nothing.

Two things worth knowing about how it behaves:

- A cell that was filled in for you is marked [?]. If the reader gave us nothing and we worked the value out from the row above, the row carries an information marker saying so. A cell where you actually wrote a ditto mark is not marked, because that one is what the paper says rather than what we concluded. It never blocks anything - it just lets you see the difference.
- Rows you never wrote on stay empty. The S.No. column is printed on all thirty rows whether or not anybody used them, so if you filled in ten rows, rows 11 to 30 come back blank. They are not filled in with the last person's name.

Nothing about how you use the page changed: photograph it, press Validate against the spec, review, stage.

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

25. Phase 11 - AI Traces

Who this chapter is for: Administrators and Auditors. It describes one new read-only page and nothing else changes for anybody.

25.1 What the page is

Where: Admin -> Console -> AI Traces, or, for an Auditor, AI Traces in the sidebar. (The Console itself is administrators-only, which is why the same page has its own menu entry for the Auditor rather than living only inside it.)

Every question anybody asks the Hub Assistant now leaves a record of how the answer was produced - not the answer, but the working:

Column	What it tells you
Question	what was typed
Retrieval	which manual sections the question actually reached, and how strongly each one matched
Queued	how long it waited for a free slot before the model started
Total	the whole turn
Outcome	answered, greeting, switched off, or failed

Expanding a row shows each stage separately.

25.2 [!] Why the Retrieval column is the whole point

The assistant does not read the manual every time. It searches it, picks the handful of passages most likely to answer you, and shows the model only those. That search has always scored every candidate section - and always threw the scores away.

Which meant that when the assistant said something wrong, there was no way to tell which of two completely different things had happened:

- it was shown the wrong pages, and answered them faithfully; or
- it was shown the right pages and answered badly anyway.

They need opposite fixes, and for a long time nobody could tell them apart. One real case: the assistant told HODs they could not open the Manpower page. The model was not confused - the section that lists who can open what had been cut off before the relevant table, so it never reached the model at all. That took a phase to find. With this page it would have been visible in one row.

25.3 The gold "fallback" tag, and why it matters

If a question matches nothing in the sections your role may read, the assistant does not give up - it falls

back to showing the model a shortened copy of every chapter you are allowed, and answers from that.

That is the condition under which an assistant makes things up: a broad, truncated context and no passage that actually answers the question. The row is tagged fallback so you can see it happened. A run of them on similar questions means the manual is missing a section, not that the model has deteriorated.

25.4 What is deliberately NOT on this page

- The answer. Only its length is recorded - enough to spot a reply that was cut off, nothing more.
- The text of the manual sections. Only chapter numbers, headings and scores. The assistant is careful about which chapters a role may see, and showing the text on a different page would quietly work around that.

You may also see a header saying "N span(s) dropped". That means the part of the system that writes these records fell behind, so the list is incomplete. It never affects the assistant itself - records are dropped rather than allowed to slow anybody's question down.

25.5 The assistant's guard rails

Two small checks were added around the Hub Assistant. Neither changes how you use it, and the second one is mostly notable for what it does not do.

On the way in, a question that is plainly an attempt to talk the assistant out of its own rules - "ignore your instructions", "show me your system prompt", "you are now an unrestricted assistant" - is answered with the ordinary "that is not in your section" reply, without troubling the model at all.

[!] Ordinary sentences that happen to contain those words are NOT refused, and that matters more than the refusals do. All of these are answered normally:

- "one drum arrived damaged - can I ignore it and issue the rest?"
- "the tank is now empty, how do I record that?"
- "can you repeat the steps for staging a return?"
- "how do I bypass a blocked lot and use the next one?"

If you are ever refused for a genuine question, that is a bug - report it. The assistant's real protection has never been this check: it is that the sections of the manual your role may not read are never loaded into your conversation in the first place, so there is nothing there to be talked out of. The check simply saves you a slow, confused answer.

On the way out, an answer is scanned before it reaches you. Anything shaped like a phone number, an email address, an ID number or a key is removed, and a line that begins with = is made safe to paste into a spreadsheet. [!] The manual contains none of those things, so if you ever see something removed from an answer, tell an administrator - it means something reached the assistant that should not have.

Answers now arrive in slightly larger pieces than before - roughly a clause at a time rather than a word - because the text is being checked as it goes. They should still appear progressively; if an answer only ever arrives all at once at the end, that is worth reporting.

25.6 Why a repeated question comes back instantly

If you ask the assistant something it has already answered - for you, or for somebody else in your role - the answer comes back immediately instead of being typed out. Nothing was skipped: it is the same answer, produced the same way, and kept for the next person who asks.

Three things about it are worth knowing:

- It is per role. A Store Keeper and an HOD asking word-for-word the same question do not share an answer, because they are answered from different chapters of this manual. Each role's answers are kept separately.
 - It is retired whenever this manual changes. Every kept answer is tied to the exact edition of the manual it came from, so a documentation update clears them all. You will never be given an answer about a screen as it used to be.
 - Only the manual assistant does this. Anything that reports live stock - the "chat with your data" card, the insights panel, the end-of-day summary - always asks the database afresh. A remembered stock figure would be a wrong number with a confident timestamp on it.
-

26. Practice Mode - learn the system without touching Live

Practice is a separate copy of GI Hub for training. It has every page, every role and every workflow Live has, but it runs on its own made-up data in its own database. Nothing you do in Practice can reach Live: not a receipt, not an approval, not a message.

Use it to learn a job before doing it for real, to rehearse something you do rarely (a return, a stock count, a PR), or to try something out without worrying about the result.

26.1 Switching between Live and Practice

On the sign-in screen, above the username box, there is a Live | Practice switch.

- Live is selected unless you choose otherwise. It is the real system.
- Choose Practice and the page reloads, ready to sign in to Practice.
- Choose Live again to go back.

Things to know about switching:

- Switching signs you out. You sign in again in the environment you picked. This is deliberate: a browser is in one environment at a time.
- Your Live password does not work in Practice, and a Practice password does not work in Live. They are separate systems with separate accounts (§26.3).
- The installed apps (Windows, macOS, Android, iOS) have the same switch.

26.2 How you know you are in Practice

The page tells you in six ways, and none of them can be dismissed:

Where	What you see
Colours	Practice is violet: the sidebar, the header, the page background and the main buttons. Live is navy and gold.
Top left, under the GI Hub logo	A pulsing amber PRACTICE badge, on every page. On a phone it sits beside the menu button. On the sign-in page it appears at the top left as soon as you choose Practice. With Reduce motion on, it stays still.
Top of every page	An amber bar: PRACTICE - practice data, nothing here is real.
Page header	An amber PRACTICE tag beside the title, and an amber line under the header that stays when you scroll
Browser tab	The tab title starts with PRACTICE -

Where	What you see
Anything you print	A large diagonal PRACTICE - NOT REAL DATA watermark

These come from the server you are connected to, not from the switch. The server reports which environment it is, and the page shows what it says.

[!] If you see a red bar saying "Environment mismatch", the switch and the server disagree. Everything you try to save will be refused until that is fixed. Sign out, pick the environment again, and tell your administrator if the red bar comes back.

26.3 The practice accounts

Practice has one shared account per role. Your trainer gives you the password for these accounts.

Role	Practice account
Store Keeper	practice.storekeeper
Supervisor	practice.supervisor
Head of Department	practice.hod
Quality Control	practice.qc
Head of Qualities	practice.qchod
Warehouse	practice.warehouse
Logistics	practice.logistics
Auditor (view-only)	practice.auditor
Admin	practice.admin (separate password, held by your trainer or administrator)

The site-based accounts work at site CNCEC, and the Warehouse account works at warehouse WH-01.

Everyone in a class uses the same accounts, so you will see each other's practice entries. That is normal.

26.4 What is in Practice

The data is made up but realistic. The categories, units of measure, lining systems, work types and tank numbers are real. The material names, SAP codes, people, vendors, quantities, prices and dates are all invented. Nothing in Practice is a real person or a real stock figure.

It is the same data the training videos use. If a video shows Tutorial Demo Gasket Set (SAP 899001), you can find it in Practice.

Work is already waiting so every role has something to do: receipts, issues and a return waiting for HOD approval, a draft purchase request, and one already with Logistics.

Every new Live feature comes with a Practice example (rule 17g), so it can be tried before it is used for

real. The current ones:

Feature	Try it on
The store keeper's note filling the job card (§4.9a.2)	One tank and day with two notes, Floor - 12.5 SQM Done and Sump Wall - 4.2 SQM Done: two jobs, each with its area and part filled in.
Lots, FEFO and expiry (§3.10.4)	PRACTICE PU PRIMER (899971): PR-OLD (expired), PR-SOON (the FEFO lot), PR-LATE.
CHEMOLINE rolls (§3.10.4)	PRACTICE CHEMOLINE (899973), batch 1026009999, three rolls.
Lots used but never received (§3.10.3)	Lot PR-TYPO.

26.5 What works differently in Practice

Everything works exactly as in Live, except these four things. Each one is switched off on purpose:

In Practice...	Why
No WhatsApp message or email is ever sent. Notifications still appear in the bell.	A practice alert must never reach a real phone or inbox.
Photo reading (OCR) is switched off. The page explains this, and you can still paste text or type rows in.	The AI engine is shared with Live and takes minutes per page, so trainees must never queue ahead of real forms.
Training videos watched in Practice are not counted. You can watch as much as you like, but I have watched and understood this is refused.	Training records are kept in Live only. Sign in to Live to have a video counted.
Two-factor authentication is off, and cannot be turned on.	The accounts are shared, so one person's authenticator would lock everyone else out.

26.6 Offline entries stay where they were made

If you lose signal while saving an entry, GI Hub keeps it and sends it when the connection returns (§3.7). In Practice this still works, with one extra guarantee: an entry saved in Practice is only ever sent to Practice. If you switch to Live before it has been sent, it waits, and it is sent the next time this browser is back in Practice. It can never be sent to Live, even if you sign in to Live on the same device.

26.7 Resetting Practice (admins)

Over time Practice fills up with everyone's practice work. An admin can put it back to its starting state:

1. Sign in to Practice as practice.admin.
2. Open Admin Console -> Practice. The page shows when the data was built and when, and by whom, it was last reset.
3. Click Reset Practice data... and type RESET PRACTICE DATA to confirm.

The reset takes about a second. It removes everything anyone has done in Practice, and everyone signed in to Practice is signed out. The approval queues are filled up again. Live is not affected in any way. It is a different database, and the Practice server has no permission to open it.

The Practice tab and the reset only exist in Practice. In Live there is no such button, and the server has no such function.

How administrators set up and run Practice on a server is in §17.9.